

Backlink Admissibility in Cyclic Proofs for the Calculus of Inductive Constructions

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Abstract. Cyclic proofs close goals by backlinking to previously-seen configurations, discovering induction principles post-hoc rather than prescribing them upfront. We prove that backlinks are *admissible* in the Calculus of Inductive Constructions (CIC): every cyclic proof can be transformed into a standard CIC proof that uses only the native fixpoint binder (**tFix**) for recursion. The transformation is mechanised in Rocq with 0 axioms. The proof is a corollary of a contextual-equivalence theorem for a supercompiler: the residual program read back from a valid cyclic proof graph is CIU-equivalent to the source and contains no backlink constructs.

1 Introduction

Cyclic proof theory [1,2] extends standard proof systems with backlink rules that allow a premise to reference a previous goal, forming a directed cycle. The cycle is sound if it satisfies a global *trace condition*: every cycle must contain a *progress event*—a case-split on a structurally smaller term. Brotherston and Simpson prove that for first-order logic with inductive definitions, backlinks are *admissible*: every cyclic proof corresponds to a standard proof with an explicit induction rule.

This paper extends their result to the Calculus of Inductive Constructions (CIC) [3,4], the type theory underlying Rocq. The central theorem is:

Theorem 1 (Backlink admissibility). *For every valid cyclic proof (rooted pre-proof plus well-founded ranking) in our calculus, there exists a standard CIC term using only **tFix** for recursion that is contextually equivalent to the root goal.*

The proof is a corollary of the CIU soundness theorem for a supercompiler mechanised in Rocq. When the supercompiler reads back a cyclic proof graph into a residual term, it replaces backlinks with **tFix** binders. The CIU theorem proves the residual is observationally indistinguishable from the source. Hence the residual *is* the standard proof.

2 Calculus

We work with a core CIC: terms are variables, sorts, dependent products, λ -abstractions, applications, fixpoints, inductive type formers, constructors (**tRoll**), and dependent case expressions (**tCase**). Typing follows standard CIC rules.

A *configuration* is a typing judgement $\Gamma \vdash t : A$. A *configuration graph* is a finite directed graph where each vertex v carries a configuration $\text{label}(v)$ and a list of successors $\text{succs}(v)$. It is a *rooted pre-proof* if every vertex satisfies a local sequent rule: $\text{label}(v)$ is the conclusion of a rule whose premises are the labels of its successors.

3 Cyclic proofs

A *cyclic backlink* closes a goal by pointing to a previously-seen vertex whose configuration is an instance of the current goal under an explicit substitution. A pre-proof becomes a *cyclic proof* when it satisfies a global progress condition: every directed cycle must contain at least one *progress edge*—a case-split on a neutral scrutinee. This guarantees that the recursion is well-founded.

The progress condition is enforced by a *budget trace* construction [1]: vertices are lifted to pairs (v, k) where $k \in [0, |V|]$. Progress edges consume one unit of budget ($k \mapsto k - 1$); non-progress edges preserve it ($k \mapsto k$). Since every cycle contains a progress edge, the lifted graph is acyclic.

4 Residualisation eliminates backlinks

The backlink rule appears only in the cyclic proof graph. When we *residualise*—read back the graph into a CIC term—backlinks are replaced by **tFix** binders. Specifically:

- A vertex with no successors (a leaf) residualises to its own term.
- A vertex with one successor residualises by applying the driving step (unfold, β -reduce, apply constructor, etc.) to the successor’s residual.
- A vertex that is the target of a backlink residualises to a **tFix** whose body is the residual of the vertex’s successors, and whose recursive call sites reference the **tFix** binder.

The result is a well-typed CIC term that uses only standard constructors (**tLam**, **tApp**, **tFix**, **tCase**, **tRoll**). No backlink constructs remain.

5 CIU equivalence

The soundness guarantee is *CIU equivalence* (contextual equivalence under all closing substitutions):

$$t \approx_{\text{ciu}} u \equiv \forall \sigma, v. (t[\sigma] \Downarrow v \iff u[\sigma] \Downarrow v)$$

The mechanised theorem (`supercompile_ciu_soundness_untyped`) establishes:

Theorem 2 (CIU soundness). *For any environment Σ , fuels f_s, f_r , context Γ , term t , and type A , if the cyclic proof construction succeeds with the trace condition,*

$$SC.supercompile_jTy_tc\ f_s\ \Sigma\ \Gamma\ t\ A = \text{Some}(v, b)$$

then the residualised term is CIU-equivalent to the source:

$$t \approx_{\text{ciu}} SC.residualise_cfg\ f_r\ \Sigma\ b\ v\ 0\ \emptyset.$$

The proof (mechanised in Rocq, 0 axioms) is by fuel induction on the residualiser, with the trace condition excluding non-terminating self-loops. Congruence lemmas lift CIU through all nine term constructors.

6 Admissibility

Theorem 3 (Backlink admissibility). $\forall\ \Sigma, f_s, f_r, \Gamma, t, A,$

$$SC.supercompile_jTy_tc\ f_s\ \Sigma\ \Gamma\ t\ A = \text{Some}(v, b) \implies \exists\ t_{\text{std}},\ t \approx_{\text{ciu}} t_{\text{std}} \wedge \text{is_standard_cic}(t_{\text{std}}).$$

In words: every valid cyclic proof (a supercompilation run that passes the trace condition) yields a standard CIC term using only native constructors (`tLam`, `tApp`, `tFix`, `tCase`, `tRoll`) that is observationally indistinguishable from the source.

Proof. Take $t_{\text{std}} = SC.residualise_cfg\ f_r\ \Sigma\ b\ v\ 0\ \emptyset$. The residual consists only of standard CIC constructors by construction (Section ??). CIU equivalence follows from Theorem 2. $\square$

The mechanised proof (`BacklinkAdmissible.v`) is a four-line corollary: extract the residual via `supercompile_jTy_tc`, apply `supercompile_ciu_soundness_untyped`, reflexivity.

7 Discussion

The key observation is that the admissibility transformation is not a separate construction—it is the same residualisation that the supercompiler already performs. The CIU theorem proves equivalence; the residualiser eliminates backlinks; the result follows trivially.

Compared to Brotherston and Simpson’s admissibility result for first-order logic, our proof is simpler because (a) the cyclic proof graph is constructed by a deterministic supercompiler rather than an arbitrary proof search, and (b) the residualisation function explicitly replaces backlinks with `tFix`, making the transformation syntactic rather than semantic.

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References

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